4 Lecture 4 (January 17, 2025): Representations as comodules

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4.1 The representation–comodule dictionary

Let G be an affine group scheme over a field k, and put $A = k[G]$. We prove Theorem 3.8.

Proof. Given a right coaction $\rho(v) = \sum v_0 \otimes v_1$, define, for $g : A \to R$,

$$r_R(g)(v \otimes a) = \sum v_0 \otimes g(v_1)a.$$

This map is R-linear. Coassociativity and the definition of the product in $G(R)$ give $r_R(g)r_R(h) = r_R(gh)$; counitality gives $r_R(e) = \mathrm{id}$. In particular $r_R(g^{-1})$ is its inverse. Composition of algebra maps $R \to R'$ proves naturality.

Conversely, suppose r_R is such a natural action. Evaluate at the universal point $u = \mathrm{id}_A \in G(A)$ and define

$$\rho(v) = r_A(u)(v \otimes 1).$$

For every $g : A \to R$, naturality gives $r_R(g)(v \otimes 1) = (\mathrm{id}_V \otimes g)\rho(v)$, so this coaction, if valid, recovers the action. Over $A \otimes A$, the two points $\iota_l(a) = a \otimes 1$ and $\iota_r(a) = 1 \otimes a$ multiply to Δ. Applying the action law and naturality to this identity gives $(\rho \otimes \mathrm{id})\rho = (\mathrm{id} \otimes \Delta)\rho$. Evaluating at $\epsilon : A \to k$, the identity of $G(k)$, gives counitality. The displayed formulas show that the constructions are inverse. Notice that $u = \mathrm{id}_A$ is the universal point, not generally the identity point $u_A\epsilon$ of $G(A)$. ∎

4.1 Definition. A morphism of right A-comodules $f : M \to N$ is a linear map satisfying

$$\rho_N f = (f \otimes \mathrm{id}_A)\rho_M.$$

By the formulas above, this is equivalent to $G(R)$-equivariance after extension of scalars to every R.

4.2 Example. The *regular comodule* is $A \xrightarrow{\Delta} A \otimes A$. The *trivial comodule* is $k \to k \otimes A$, $1 \mapsto 1 \otimes 1$; it uses the unit $k \to A$, not the counit $A \to k$.

4.2 The abelian category of comodules

The next constructions require only that A be a coalgebra over k. A subspace $N \subseteq M$ is a *subcomodule* if $\rho_M(N) \subseteq N \otimes A$. The quotient then has the well-defined coaction

$$\overline{\rho}(m + N) = \sum (m_0 + N) \otimes m_1.$$

Coassociativity and counitality follow from those on M.

4.3 Theorem. *The category of right A-comodules is a k-linear abelian category. Kernels, cokernels, images and direct sums are computed on the underlying vector spaces. The same holds for the full subcategory of finite-dimensional comodules, with finite direct sums. The forgetful functors are exact.*

Proof. For a comodule map $f : M \to N$, the identity $(f \otimes \mathrm{id}_A)\rho_M = \rho_N f$ and exactness of tensoring over a field give

$$\rho_M(\ker f) \subseteq \ker(f \otimes \mathrm{id}_A) = \ker(f) \otimes A.$$

The image is also a subcomodule, and the ordinary quotient $N/\operatorname{im} f$ inherits a coaction. These constructions have the kernel and cokernel universal properties because their factorizations are linear maps whose coaction identities descend from that of f. Moreover the vector-space isomorphism $M/\ker f \simeq \operatorname{im} f$ respects coactions.

The direct sum has componentwise coaction. For a finite sum $M_1 \oplus M_2$, the usual inclusions i_j and projections p_j are comodule maps and satisfy

$$p_j i_j = \mathrm{id}_{M_j}, \qquad p_k i_j = 0 \ (k \neq j), \qquad i_1 p_1 + i_2 p_2 = \mathrm{id}_{M_1 \oplus M_2}\,.$$

The zero object and these biproducts make the category additive; comodule maps form linear subspaces of ordinary Hom spaces. The kernel, cokernel and image–coimage statements therefore prove that it is abelian. All these constructions preserve finite-dimensionality when the inputs are finite-dimensional. ∎

4.3 Local finiteness

4.4 Theorem. *Every element of an A-comodule belongs to a finite-dimensional subcomodule. Consequently every comodule is the directed union of its finite-dimensional subcomodules.*

Proof. Write $\rho(m) = \sum_{i=1}^{r} m_i \otimes a_i$ with the a_i linearly independent, and let $N = \operatorname{span}\{m_1, \ldots, m_r\}$. Choose linear functionals $\lambda_j \in A^*$ with $\lambda_j(a_i) = \delta_{ij}$. Coassociativity gives

$$\sum_i \rho(m_i) \otimes a_i = \sum_i m_i \otimes \Delta(a_i).$$

Applying $\mathrm{id}_M \otimes \mathrm{id}_A \otimes \lambda_j$ shows $\rho(m_j) \in N \otimes A$. Counitality gives $m = \sum_i \epsilon(a_i) m_i \in N$. Thus N is the required subcomodule. Finite sums of finite-dimensional subcomodules are finite-dimensional, which proves the directed-union assertion. ∎

4.5 Definition. An affine group scheme is *algebraic* if its coordinate algebra is finitely generated as a k-algebra.

4.6 Corollary. *Every affine group scheme over k is an inverse limit of algebraic affine group schemes.*

Proof. It suffices to show that a commutative Hopf algebra A is the filtered union of its finitely generated Hopf subalgebras. Given $a \in A$, choose a finite-dimensional right subcomodule $V \subseteq A$ containing a. In a basis $v_1, \dots, v_n$ write $\Delta(v_j) = \sum_i v_i \otimes c_{ij}$. The coaction identities give

$$\Delta(c_{ij}) = \sum_l c_{il} \otimes c_{lj}, \qquad \epsilon(c_{ij}) = \delta_{ij}.$$

Applying $\epsilon \otimes \mathrm{id}$ to $\Delta(v_j)$ shows that every v_j is a linear combination of the c_{ij}. Hence the algebra generated by the c_{ij} and $S(c_{ij})$ contains a. It is stable under comultiplication and counit, and also under S, since $S^2 = \mathrm{id}$ in a commutative Hopf algebra (inversion squared is the identity). It is therefore a finitely generated Hopf subalgebra. Applying the same argument to finite sets makes these subalgebras a directed family. Taking spectra reverses the resulting filtered union to an inverse limit. ∎

The subalgebras here are finitely generated as algebras, not necessarily finite-dimensional as vector spaces; $k[x]$ already illustrates the distinction.

4.4 Tensor products

For any bialgebra A, the tensor product of right comodules has coaction

$$\rho_{M\otimes N}(m \otimes n) = \sum m_0 \otimes n_0 \otimes m_1 n_1.$$

Equivalently, apply $\rho_M \otimes \rho_N$, interchange the middle factors using $\tau : A \otimes N \to N \otimes A$, then multiply the two A factors. The bialgebra axioms imply the coaction identities. The unit object is the trivial comodule k, and the ordinary associativity and unit maps are comodule maps. When A is commutative, the flip $M \otimes N \to N \otimes M$ is colinear, since $m_1 n_1 = n_1 m_1$. Thus representations of affine group schemes have a symmetric monoidal structure. For a general Hopf algebra the ordinary flip need not be colinear.

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